

Ramco Institute of Technology
Department of Mechanical Engineering
Academic Year: 2020-2021 (Odd Semester)

ME8099 Robotics

Date: 26.10.2020

**Implementation of Robot in an Industry- Micro Assignment-
Decision Making**

Name of the Faculty Member: Mr. S. Valai Ganesh, AP/Mech

Let us assume that you are being the part of Automation division in an industry. Your industry would like to introduce a robot for specific applications like material handling/welding/painting/other processing applications. For this, company give a open bid tender to all suppliers. The quote is attached this assignment (as a separate file). You are the in-charge for procurement of the robot. Compare and answer the following situation and try to justify your selection.

S. No	Criteria	Supplier A	Supplier B	Supplier C
1.	Type of application			
2.	Model Name			
3.	No. of axis			
4.	Pay load/ Weight carrying capacity			
5.	Repeatability			
6.	Speed (Angular Velocity)			
7.	Reach			
8.	Working Envelope			
9.	Configuration type			
10.	Mounting type			
11.	Dimensions			
12.	Weight			
13.	Environmental conditions			
14.	Electrical requirements			
15.	Cost			
16.	Range of all joints			
17.	Others*- T & C (which is also consider for the comparison)			
18	Selection			

Justification of your selection:

Signature of the Student

1. GENERAL

Dear Sir,

We are very thankful for the opportunity you have provided to study your requirement and submit herewith our proposal for MOTOMAN robot model GP12 with YRC1000 Controller. This offer tendered for discussion only, does not constitute an offer to sell and/or term to contract and YIND_R can change without notice, make any change in YIND_R's own discretion. Any contract to sell the subject matter of the proposal shall be subject to prior mutual agreement as to price (which may be different than that shown herein), schedule, scope of work and terms"."

1. 1 Introduction

YASKAWA India Pvt, Ltd is uniquely qualified to provide the integrated Robotic Automation Systems and Services for Customer's automation system program. YASKAWA has numerous successful application experiences in the field of Factory Automation around the world.

The information and descriptions in this proposal are provided to define YASKAWA approach to developing a clear Customer understanding of YASKAWA's products and services to satisfy Customer's requested automation needs. Based on the information provided to YASKAWA, YASKAWA is confident that YASKAWA India Pvt., Ltd meet the requirements of Customer's robotic application, and has proposed the best solution available from both functional and economic standpoint.

2. Scope of Work

YASKAWA offers Robot model

- GP12 ROBOT with YRC 1000 Controller
- Robot Power Cable 4m
- Pendant with 8-meter cable
- Education Cell – 1 Set [As per the Industrial Standard, Handling)

Registered Office:

Yaskawa India Private Limited
CIN no. U29253KA2010FTC054999
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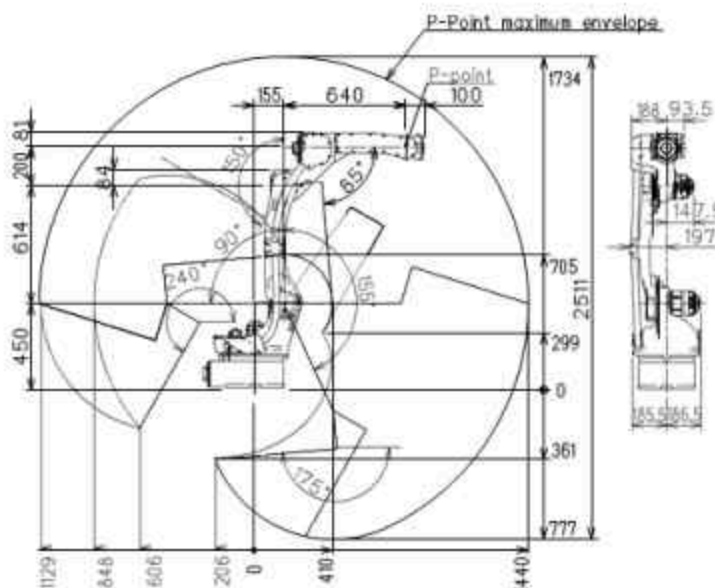
3.1 MOTOMAN ROBOT MODEL GP12 with YRC1000 Controller

Motorman's high performance, six-axis GP12 robot has a faster axis speed than conventional model. This achievement combined with light weight body and robot controller with the high accuracy path control and the vibration suppression control, reduces the residual vibration when starting up and stopping, resulting in shortened cycle times for common welding patterns.

The GP12 features a maximum reach of 1440 mm. The robot's B-Axis features an expanded range of motion, which improves circumferential movement.

Key benefits

- Greatest reach in its class: 1440 mm
- High payload capacity: 12 kg
 - Enables the mounting of heavy water-cooled torches or front-pull system
 - Specially developed for welding thin work pieces
- Hollow front arm
 - Diameter 50 mm
 - Minimized interference contours
- Streamlined design



Model	GP12	
Structure	Vertically Articulated	
Degree of freedom	6	
	Payload	12 kg
	Wrist Payload	
	On U-arm	10 kg
Maximum reach	1440 MM	
Repeatability Note3	±0.08 mm	
	S-axis	-170° - +170°
	L-axis	-90° - +155°
	U-axis	-85° - +150°
	R-axis	-200° - +200°
	B-axis	-150° - +150°
	T-axis	-455° - +455°
	S-axis	4.53 rad/s, 260°/s
	L-axis	4.01 rad/s, 230°/s
	U-axis	4.53 rad/s, 260°/s
	R-axis	8.20 rad/s, 470°/s
	B-axis	8.20 rad/s, 470°/s
	T-axis	12.2 rad/s, 700°/s
	R-axis	22 N · m (2.2 kgf · m)
	B-axis	22 N · m (2.2 kgf · m)
	T-axis	9.8 N · m (1.0 kgf · m)
	R-axis	0.65 kg · m ²
	B-axis	0.65 kg · m ²
	T-axis	0.17 kg · m ²
Approx. mass	130 kg	
Protected structure	Body: IP54, Wrist: IP67	
Paint color	YASKAWA blue	
Power requirements	1.5 kVA	
Noise Note5	Less than 76.7 dB	
Way of mounting Note6	Floor-mounted, wall-mounted, Tilt-mounted, ceiling-mounted	
Adaptive controller	YRC1000	
Manipulator cable	Standard: 4 m	

Note1: SI units are used in this table. However, gravitational unit is used in ().

Note2: Load on the U-arm will vary depending on the load weight of the wrist part.

※Please refer to "(4-1-1) Allowable Mass" for more information. Note3: Repeatability conforms to ISO 9283.

Note4: Please refer to "(3) Allowable Load for Manipulator wrist" for the allowable moment and allowable inertia.

Note5: Conformed to ISO6926

Note6: When the manipulator is installed on a slope or a wall, the motion of S-axis will be limited.

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3.1.2 YRC1000 CONTROLLER

Fast, feature packed, highly reliable and easy to use Yaskawa YRC1000 robot controller is the PC based high performance robot controller. The YRC1000 controller offers high speed processing, unmatched memory (200,000 steps, 10,000 instructions), built-in Ethernet and a robust PC-architecture. Smallest in its class, fits in wide range of layouts. Realizes compact facility with high density layout.



The YRC1000 easily handles multiple tasks and can control multiple robots plus external axes and I/O devices. Advanced Robot Motion control (ARM) provides high performance path accuracy and motion control. The soft float enables for ease in machine tending application. The control system's PC architecture offers unlimited communication systems via a wide range of communication protocols.

Item	Specification
Construction	Free-standing, enclosed type
Dimensions	598 (W) x 490 (H) x 427 (D) mm(excluding protrusions)
Mass	70 kg
Cooling system	Indirect cooling
Power Supply	Three-phase 200~240V AC (-15 to +10%) 50/60Hz (±2%) A specification of connectable power supply differs depending on the controller type.
Grounding	Three-phase 200~240V AC :Grounding resistance: 100 Ω or less, exclusive grounding*1
Digital I/O	For Japan / Asia / North America: NPN For Europe: PNP
	Specific signals (hardware): 19 inputs and 6 outputs
	General signals (standard max.): 40 inputs/40 outputs (transistor: 32 outputs, relay: 8 outputs)
Positioning system	By serial communication (absolute encoder)
Drive unit	SERVOPACK for AC servomotors
Acceleration/deceleration control system	Software servo control
Memory capacity	200,000 steps, 10,000 instructions
CIO ladder	Max. 20,000 steps
Built-in transformer	None
Safety category	Safety category 3, PL e

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Manipulator, YRC1000 robot controller, manipulator cable, programming pendant



3.1.3 PROGRAMMING PENDANT

MOTOMAN teach pendant is a user-friendly touch screen pendant with Windows CE operating system and full color LCD touch-screen display. The other features of the teach pendant is as following:

Complete access to all operations from programming to maintenance.

- Reduces programming time.
- Possibility to create your own user menus.
- Eliminates the need for separate operator's panel.

Standard Specifications of Programming Pendant

Item	Specificatio
Type	JZRCR-APP01-1
Dimensions	152 (W) x 299 (H) x 53 (D) mm (excluding protrusions)
Mass	730 g
Material	Reinforced plastics
Operation s device	Select keys Axis keys



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	Numerical/application keys
	Mode select switch with a key (mode: teach, play, and remote) Emergency stop button Enable
Display	640 x 480 pixels color LCD, touch panel (Alphanumeric characters, Chinese characters, Japanese)
IEC Protection Class	IP54
Cable length	Standard: 8m, Max: 36m (optional)
Key sheets	Arc, Spot, Material handling, Jig-less, Spot + arc

4.0 CUSTOMER RESPONSIBILITIES

4.1. UTILITIES

4.2 .1PRIMARY POWER

- Robot controller: AC 200 $\pm 10\%$, 50Hz ± 1 Hz,3 Phase +E(1.5 KVA)
- Power for other auxiliary units.
- Individual isolated Earthling for robot controller: less than 100 Ohm.
- Air Supply of Dry Air at 6.0 Bar thru FRL at each Cell.(Specified location)
- Flow of air as per YIND_R requirement.
- Line circuit breaker, power distribution box and cables for mains supply to the Control Panel of System.

4.2.2 Installation Environment

Temperature	0° to 45°C
Humidity	20 to 80% RH at constant temperature
Vibration Acceleration	Less than 4.9 m/s ² (0.5 G)
Others	Free from corrosive gas or liquid, or explosive gas. Free from water, oil, or dust. Free from excessive electrical noise (plasma).

4.2.3 SAFETY

- Customer is responsible for safe operation and maintenance of the equipment.
- Operators should wear protective head-wear while working in the operating area of the robot.
- Customer should not bypass any of the safety equipment provided with the system.
- Only trained personnel should be allowed to operate the robot.

5.0 CONDITIONS TO THE PROPOSAL

- Our proposal is based on the specifications and contents described in this document. If there are any deviations from the specifications, YIND_R reserves the right to reevaluate the deviations and submit a revised proposal.
- Customer is responsible for maintaining the quality of the input parts.
- All consumable will be in customer scope.
- Robot Installation will be on a flat concrete floor that is able to bear the dynamic loads (forces and moments) of the robot as specified by the robot manufacturer. Reinforcements to civil construction, if required, are the responsibility of the customer.
- No any Spare parts supply.
- Logic modification & Wiring on the machine side for interfacing robot will be in Customer scope.
- Site preparation including foundation, foundation bolts, etc or cable scheduling.
- Other peripherals which are not specified in the offer.
- Safety installations such as for example fire-extinguishing system not included in the offer.
- Lifting equipment at the Buyer's plant, with operator for off-loading and when needed for the erection and installation.
- Opening for delivery to installation site.
- Lockable room for storage of the high-grade components after delivery if required
- Inside plant transportation of the devices to site of installation.
- Provision of lifting devices during installation (crane or forklift, respectively, with operator) Installation force and installation works.
- Installation manpower under YIND_R supervision. Temporary assistance persons during off-loading, erection, installation and start up at the Buyer's plant.
- Repair of devices or system units, respectively, damage of which cannot be proved to have been done by YIND_R personnel or persons appointed by them.
- Provision of dry contacts for control signals outputs, inputs signals should be +24V DC are available for the peripheral equipment.
- Power supply cabling including installation material and systems communication cabling and cable ducts are not included in the Seller's scope of supplies. Other peripheral equipment has to be ready assembled and cabled and the control and interfaces have to be operative.
- All civil works before equipment installation, cleaning of plant before production starts.
- If the installation of YIND_R portion could not be achieved on time or interrupted of job due to out of the Buyer's faults, and extra working hours are requested, additional charge would be borne by the Buyer
- Guarantee/Warranty and maintaining regular spare for all Non YIND_R equipment will be in the scope of Customer itself.

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6.0 Defects Liability Period

12 months have been considered for the defects liability (warranty) period.

All kind of battery, grease & other consumable not covered in warranty.

6) COMMERCIAL OFFER :

S.no	Description	QTY	Price in INR [Ex Works Chennai]
1	Yaskawa make handling cell along with accessories. Robot Model : GP12 Handling Robot -1 set YRC1000 Robot Controller Programming pendant Robot Cable : 4m Programming pendant cable :8 m Education Cell – 1 Set [As per the Industrial Standard, Handling]	1 SET [1 Robots with Education Cell]	INR 19,50,000

NOTE: GST & TCS will be applicable extra @ Actual .

STANDARD TERMS AND CONDITIONS**A) PAYMENT TERMS:**

40% Advance Along with PO balance 50 % along with taxes against the PI before dispatch and 10 % after successful installation within 45 days of invoice .

B) DELIVERY PERIOD: 3 Months from the date of PO & Advance.**C) DELIVERY TERMS:** Door Delivery @ Ramco Institute**D) TAXES & DUTIES & INLAND TRANSPORTATION:**

Duties Inclusive on the above Price & local taxes will be applicable extra @ actual .

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E) PACKING AND FORWARDING: Inclusive

F) VALIDITY OF OFFER: 30 days from the date of this document.

G) ERECTION & COMMISSIONING:

Unless explicitly specified, the cost of the equipment does not include erection or commissioning services. Commissioning services are available at extra cost. Service tax shall be charged extra in such cases @12% +2%Cess+1%SHE.

H) WARRANTY:

YIND will provide warranty for the system / items supplied will be free from manufacturing defects for a period of 12 months from the date of Invoice. The warranty does not cover improper operation and maintenance, or accidents caused due to negligence on the part of operator.

I) CANCELLATION:

In case of cancellation of LOI / Purchase order, following cancellation charges will apply

Within 1 month of LOI / Order: 20%

Within 2 month of LOI / Order: 50%

After 2 months of order: 100%

J) LIMITATION OF LIABILITY:

In no event we shall be liable for any consequential loss or damage arising out of or connected with this Purchase order in any way whatsoever

K) JURISDICTION: Subject to Gurgaon, Haryana jurisdiction

L) FORCE MAJEURE:

Notwithstanding any other provisions of this agreement, no liability shall attach to any party, in the event of its failure to observe the terms hereof, if such failure is caused by reason of, or on account of, or in consequence of any act of GOD or act of State, plague, famine, pestilence, epidemic, fire, storm, floods, earthquake, restraints of war, mutiny, riot, civil commotion, disturbances, Government statute or

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imposition, or any other cause whatsoever beyond reasonable control of such party. We shall not be liable for damages or any other consequences in respect of delays in delivery, or nonperformance of the contract otherwise, if it is due to reasons or circumstances beyond our control, and despite due diligence and prudence

M) ARBITRATION:

Any dispute or difference arising under or in connection with this agreement or any breach there of, which cannot be settled by friendly negotiation and agreement among the parties, shall be finally settled by arbitration, with three arbitrators to be appointed, one arbitrator to be appointed by YIND, one arbitrator to be appointed by the other party and the umpire to be appointed by the two arbitrators, the arbitration being conducted in accordance with the provisions of the Arbitration and Conciliation Act, 1996. The venue of the arbitration proceedings shall be in India. The costs of arbitration will be equally shared by the parties

N)Information Notice on Security Trade Control:

This is to inform that the goods / material mentioned in this proposal, if sold to you, will not be used in relation to nuclear, biological or chemical weapons, or missiles capable of delivering these weapons. Also confirm that the goods will not be re-exported or sold to a third party who is known or suspected to be involved in relation to nuclear, biological or chemical weapons, or missiles capable of delivering these weapons, or to any sanctioned entities.

That any re-export or sale to third party, is carried out in compliance with the originating/supplying and receiving countries' export control laws, as applicable.

O) Important notification on TCS :

The Government of India has introduced Section 206C(1H) of Income Tax Act, 1961 with regard to "Tax collection at Source (TCS)". It applicable to all sellers whose turnover is more than INR 10 crores in last Financial Year.

If your purchases from us exceeds more than Rs, 50 lacs during current Financial Year, then we need to add TCS on your bill. Rates of TCS is 0.075% till March 31, 2020 & increase to 1% thereafter if you have deposited PAN No. to us.

We will deposit your TCS into Government treasury on receipt basis. This will be applicable on all invoicing from 01st Oct - 2020.

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About EMSOL SYSTEMS

Emsol Systems is a professionally managed Embedded based Product Development company specialized in implementing the trainer kits and Research & Customized lab Solutions and end Products for both Educational & Industries in the field of Electronics, Electrical & Mechanical, Instrumentation, Mechatronics, Bio Medical, Also implementing security & fire projects on turnkey basis since 2012. Our separate business unit handles distribution business of globally renowned security infrastructure system products and passive networking products.

Emsol Systems Quality is paramount and excellence is guaranteed. We doesn't deal in average or subpar products. We only sell the highest quality, most original products. We understand that original products are the driving force behind all successful Electronics products. For that reason, all of the products we sell are manufactured by our professional and also reputed firms subject to authentication using the Internet's premiere prevention tool. Our people are developing innovative technologies and creating market-specific services year after year. Emsol Systems is a multicultural and global business that believes as strongly in the latest thinking as it does the latest technology. As part of our team, you could enjoy the best of both worlds while leaving your legacy.

Emsol offering the Engineering Labs Solutions for

Engineering Labs	Hardware Requirement
VLSI Design	Xilinx Spartan 3E, Altera Cyclone II
Circuit Simulation	SPICE Software
Antenna Design	L & S Band Horn Antenna Setup with Nw Analyser
Electronics Lab	Digi. & Anag. Trainer, Power Supply, FG,CRO,IC Tester
Communication Lab	Basic, Analog, Digital & Advance Trainer
Optical Lab	Basic,Advance,Laser optical Trainer
Microwave Lab	Klystron & Gun based Setup
DSP Lab	Texas TMS320 Series Products
Embedded Systems	Micro Processor/Controller/Interface cards
Instrumentation & Control	PID, Potentiometer, Temperature,Relay,DC Motor trainer
Power Electronics Lab	Induction,BLDC,SRM,Synchronous based load setup
Mechatronics Lab	Hydraulic & Pneumatic based setup with Software
R & D Labs	Robotics, Wireless, Image processing & Cloud Computing

Technology Overview

Today, embedded engineering is considered as heart of all the technical rich products and is becoming more and more prominent in our next generation products and further, robotics / embedded industry will contributed significant inputs to the development of other industry sectors like automobile, aerospace, energy, industrial automation, consumer and service products

As most statistics say what IT was in 90's is where robotics / embedded systems stands now. And also, there is expected to be a growth of around 90% software developers by 2018, when compared with today's 2.75 million software professionals. This is a clear sign of huge demand of engineers with practical knowledge of embedded system / computational concepts. India is now looking at an electronics manufacturing industry the size of \$400 billion by 2020, from \$70 billion now*. It is time for all the young engineering to en-cash this opportunity and be part of next big technology. The future is bright for India with it being considered to be the next embedded systems hub in the world.

To bridge the gap between these industrial requirements and the present academic content, our hands-on-sessions are structured to introduce the student to the core of embedded systems, robotics and computation concepts from scratch and encourage students to create their own projects. It starts out with the very basics of electronics and programming, and goes on to prepare an objective robot using the provided kit. These sessions help students to the work on intricacies and challenges involved in solving real world problems. 70% of our training programs are based on hands-on-sessions which guarantee that each participant gets required amount of exposure to head start his career with robotics/ embedded engineering which is rapidly growing and contributing substantial input to India's growth

Key Features

- 70% of hands on learning.
- Well researched material.
- Comprehensive training with practical approach.
- Increases Employability Skills.
- Addresses Key Life Skills.
- Economical.
- Industry Recognition.
- Exposure to the Global industry workforce.
- Hands on experience on industry standard tools, processes and systems

Benefits:

- Platform to get exposed to multiple disciplines of Engineering
- Students get trained on state of art embedded technology.
- It serves as a brilliant exposure to help you identify your field of Interest.
- Helps in identifying your specialization for high studies.
- You get a Certification of appreciation which gives a better edge among others.
- Instead of relying on Enable them to think and build their own projects during academics.
- For meritorious candidates, industries offer internship/ scholarship during project
- Lays a path for bright career in embedded systems / application development / mechanical / robotics / etc.

Program Overview



What College Gets?

✓ In college research facilities

Universities/Colleges equipped with Robotics Center of Excellence will have access to various robots, software & hardware tools. This will help universities to create a eco system where high end research can be executed at par with other reputed universities globally.

✓ Provide Training to your Faculty/Engineer by Emsol

It is observed that advanced practical learning labs in many universities/Colleges are ideal and are NOT being actively used by students as there is a lack of skill set in the campus to train students. To overcome this hurdle for universities, Emsol has come up with a service offering to deploy a well trained robotics engineer at your campus for training your faculty until he/she will be equipped with complete knowledge to transfer it back to their students.

✓ Quick return of investment on facilities

Student/ Industry community now understands the importance of practical education. It is always observed that students prefer to study in a campus where there is good research infrastructure and hybrid cultural ecosystem. Establishing Robotics Center of Excellence will push any university a place ahead in getting new admissions and getting international exposure

✓ Free maintenance of facilities

Emsol will take the responsibility of maintaining of all robots and equipment installed in Robotics - C.O.E without any financial/technical burden to the concerned university

✓ Revenue benefits on infrastructure

Emsol will organize practical learning courses on robotics and automation, revenue generated from such courses will be partly shared with university, on an account of using the infrastructure facilities from the concerned campus.

✓ Industry Academia Interaction

Universities/Colleges when equipped with advanced facilities like Robotics - Center of Excellence and when offered to students with industry related practical learning courses; that is the best any educational institution can offer to their students. This culture will also enhance the industry academia interaction and will encourage students to be industry ready by the time they graduate

What Students Get?

✓ Certified Hands on Training Programs

Students will get access to well defined curriculum where they learn and experience advanced technologies in a practical approach using robots. This learning pedagogy will enhance the students creative, logical, analytical, problem solving, engineering and technology skills. This will help students inculcate in them the key life skills required for an engineer.

✓ Integrated with current semesters

With support from university/Colleges, all the courses in connection with Robotics Center of Excellence can be executed in-line with academics. This does NOT leave any additional burden to the students but will keep them self motivated to achieve higher in academics. Unlike other students who go for additional certified courses later after graduation, students can now get certified with required industrial technologies very much during their academics itself.

✓ Life cycle of product development, Marketing and Branding

Engineering studies is one of the junctions in our life, where we decide to continue our career in either of the administration / engineering fields. So at this stage, it is important for the student to be aware of basic life cycle of product development, marketing and branding procedures. Such eco system will give a better insight to student to select the right path and succeed progressively. As part of the course, students will be exposed such eco-system which will give them better edge over others

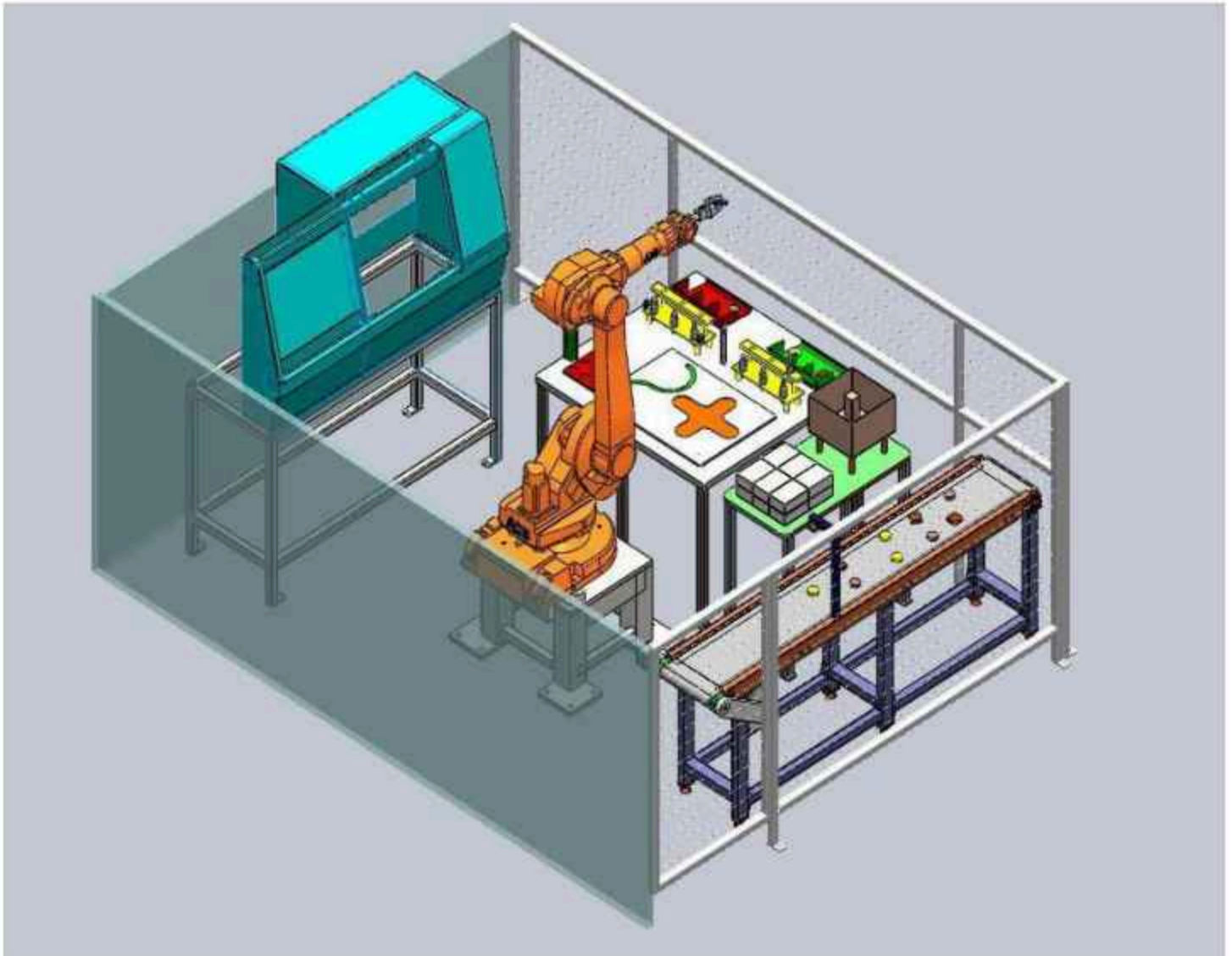
✓ Industry Connect Program

Every students who is part of our training program will have an access to Industry Connect Program, where Industries and Students share a common platform. Student will get an opportunity to get interviewed with top companies and Industries will get an access to graduates with practical knowledge and industry attitude

✓ Gain self confidence to execute innovative projects

It is very much a known fact that, most of the engineering students tend to purchase and submit their final major/mini projects, just because students have got NO interest(or) because students having NO access to required infrastructure/ technical guidance to execute the projects. In either of these cases, it is very much important for both university and student to understand that, one very important phase of engineering is the opportunity to show case our skills in the form of our final projects. When that very important phase itself is NOT handled by students properly then we @ EMSOL feel that the engineering education is of NO importance. It is to address this very important problem in education today, we have come up with our offering of Robotics Center of Excellence, where; students will have access to infrastructure and technical guidance will be provided either by university or by Emsol. Moreover students when enroll with our courses they get self-motivated and will be more than willing to execute their own projects.

Robotics & Industrial Automation Lab Layout



The above image shows the robotic cell layout. All experiments are done inside the cell. There is a limit switch connected to the door. If door opens the entire operation will be stopped. After the door gets closed, the operation will resume

LIST OF EXPERIMENTS

Exp No 1: Industrial Robot Introduction & Basic Interface Programming



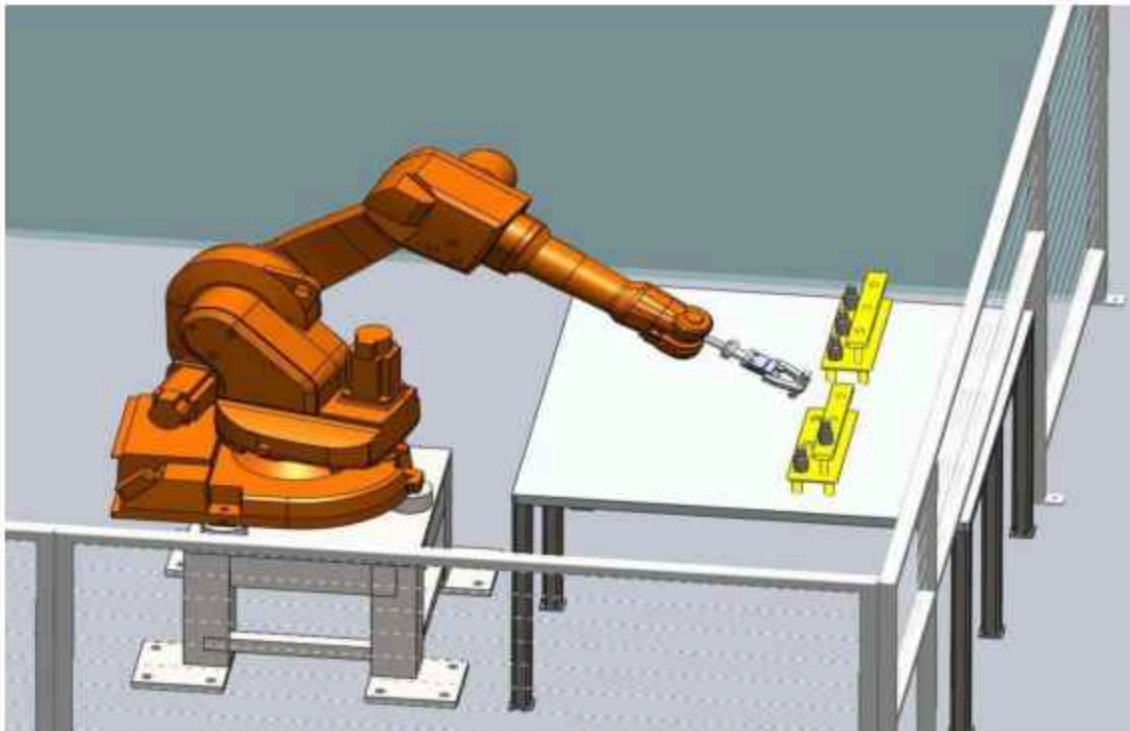
- ✓ To know about basic kinematic configurations & components of robots.
- ✓ Robot specification (payload, reach, supplementary load, speed).
- ✓ Description of controller.
- ✓ Description of end effector.
- ✓ Difference between robot working in auto mode & manual mode
- ✓ Study of KAREL programming
- ✓ Labview & its basics
- ✓ Interface with external I/Os
- ✓ Communication between robot & robot Guide:
- ✓ Hard wire communication between robot & PLC
- ✓ Communication between computer & PLC
- ✓ Programming robot via PC
- ✓ Communication & interlocks between robot PLC & CNC
- ✓ Jogging robot in axis mode
- ✓ Robot interpolation
- ✓ Jogging robot in tool coordinate system
- ✓ Work object explanation
- ✓ TCP teaching to robot

Exp No 2: PLC programming:



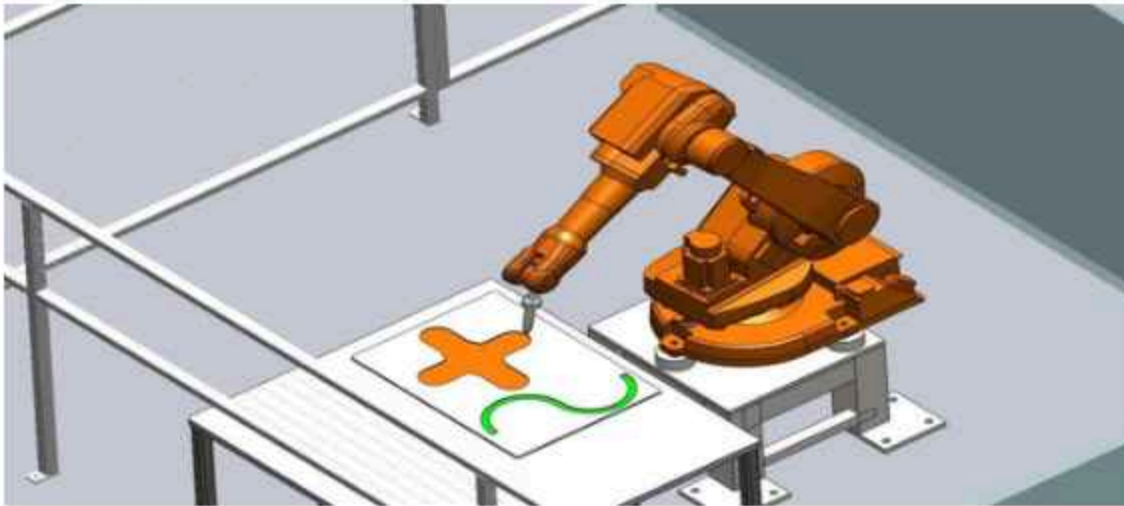
- ✓ Definition of PLC.
- ✓ Type of wiring (relay type, transistor type).
- ✓ Mode of communication with PLC.
- ✓ Type of programming.

Exp No 3: Gripper based pick and place



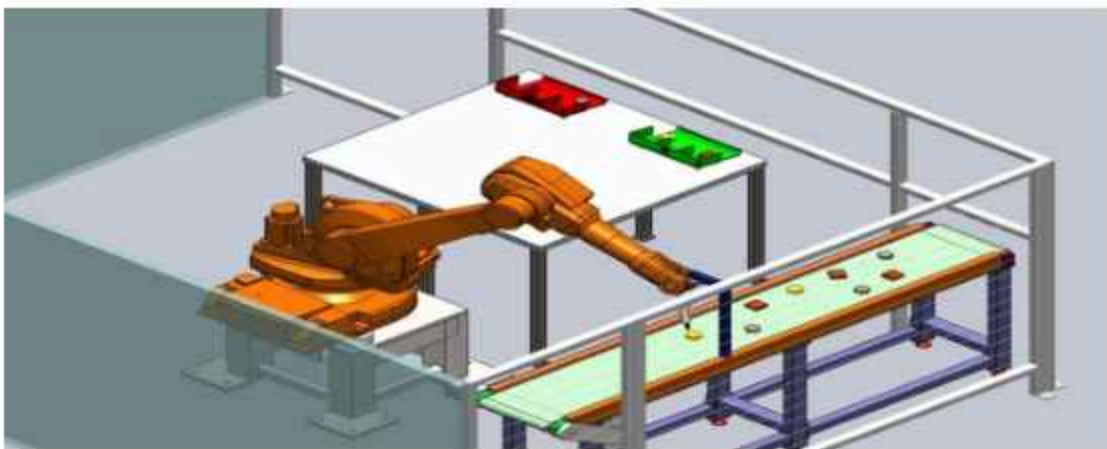
- ✓ The Components will be loaded in the pallet present in the table.
- ✓ The mechanical Gripper is attached to the arm of the robot.
- ✓ As per predefined program the robot picks the component by using the mechanical gripper from the first pallet.
- ✓ Then it places in the next pallet.

Exp No 4 : Simulated welding:



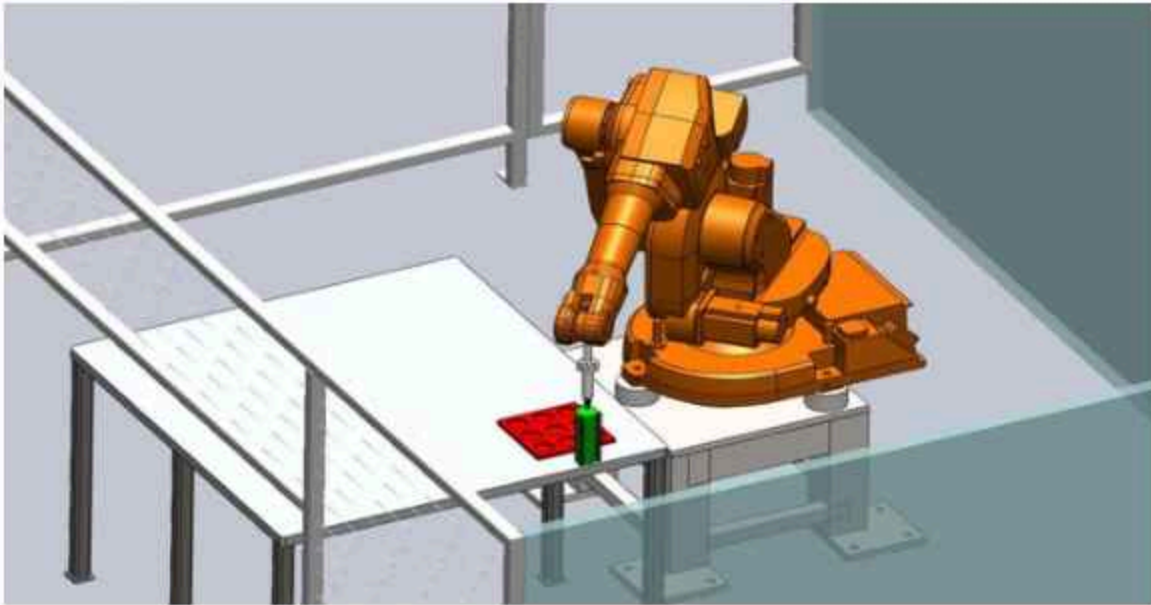
- ✓ A pre-designed path will be placed in the table.
- ✓ A dummy welding torch is attached to the arm of the robot.
- ✓ By using the predefined program the robot will identify the welding location.
- ✓ Linear welding: Robot will be programmed to weld linearly along linear path.
- ✓ Arc welding: Robot will be programmed to weld circularly along circular path

Exp No 5 : Vision inspection & sorting:



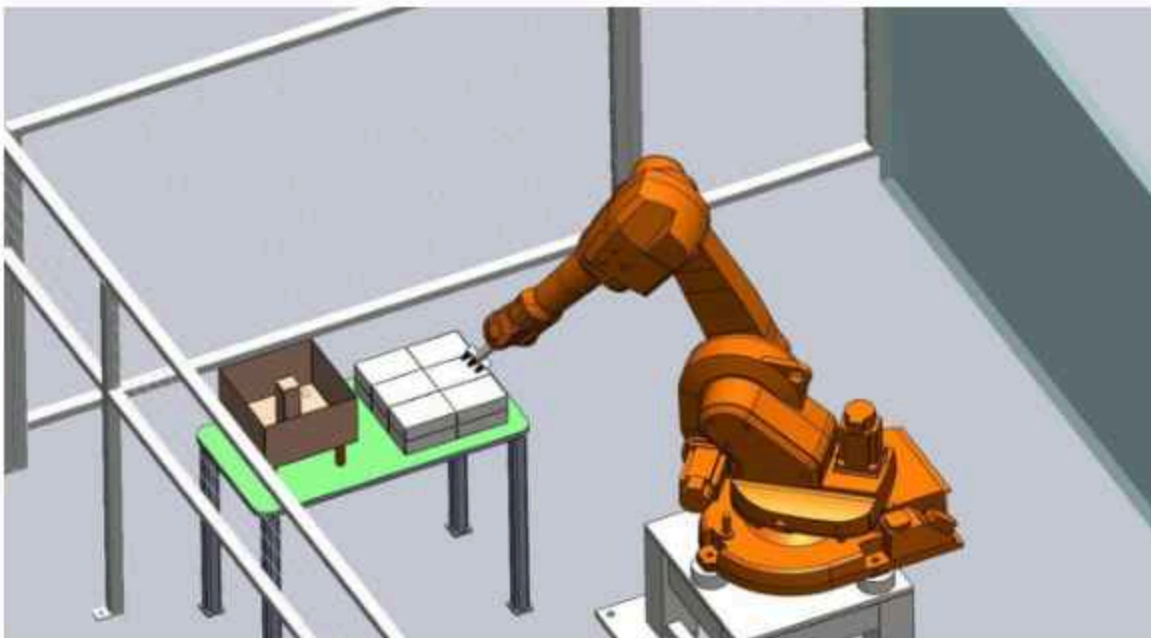
- ✓ Different shaped components (square, circle) with a combination of perfect & defective pieces will be placed in the conveyor.
- ✓ The component location would be identified by using the camera & lab view inspects the condition of the component based on image input(perfect or defective).
- ✓ By using the predefined program if the component is defective, robot picks those components using suction cup & places in the red bin
- ✓ If the component is perfect the robot picks & place those components in the green bin.

Exp No 6 : Matrix palletising:



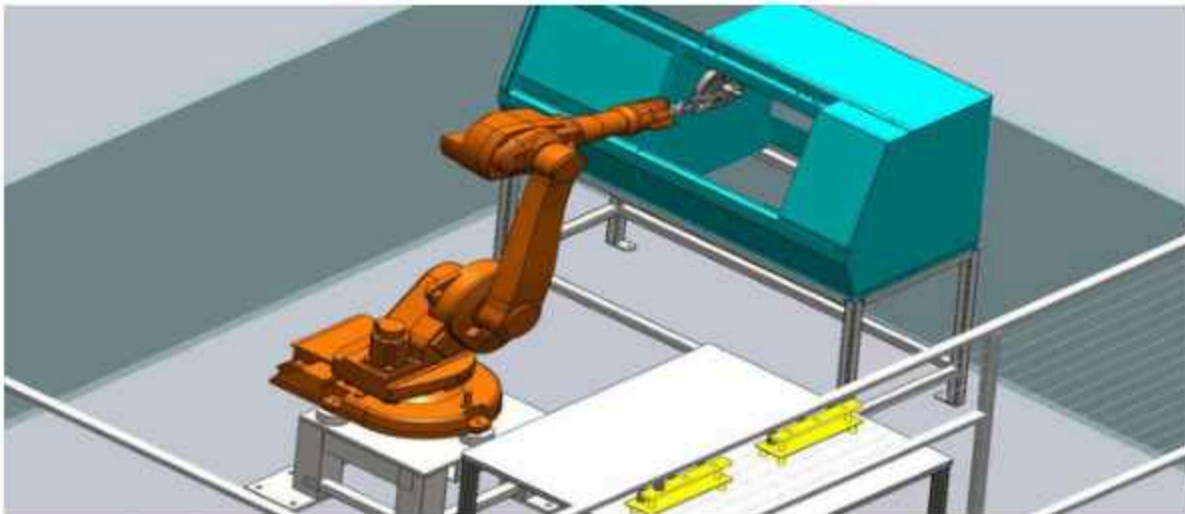
- ✓ The pallet (matrix form) & components are placed in the table.
- ✓ Suction cup will be attached to robot arm.
- ✓ Using predefined program the robot picks the each component by using the suction cup.
- ✓ Then the robot places the component in the pallet.

Exp No 7 : Carton palletising:



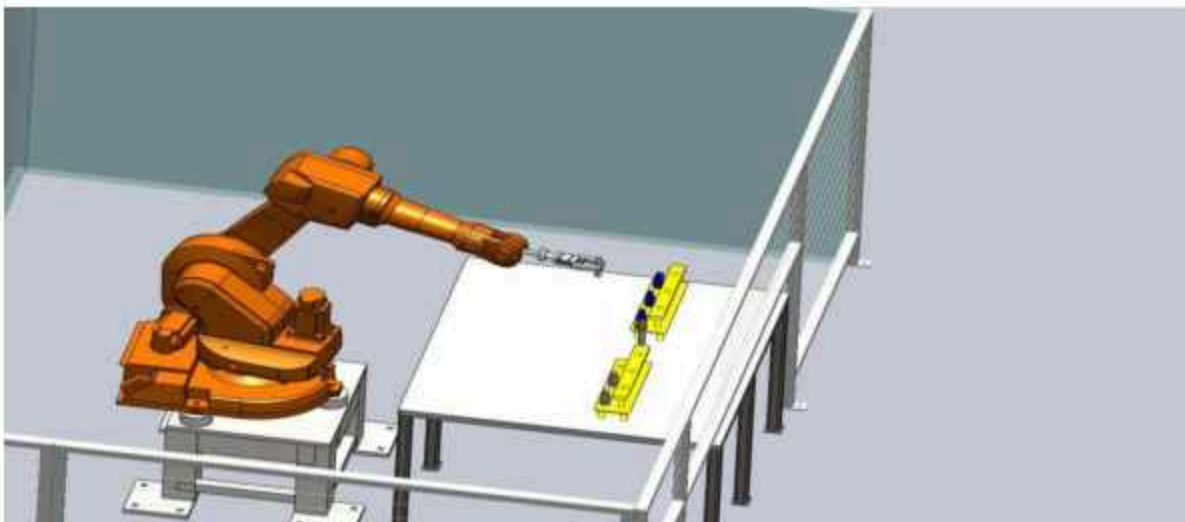
- ✓ Wooden boxes & palletising box will be placed in the table
- ✓ The vacuum gripper will be attached to the robot arm.
- ✓ By using the suction cups the wooden boxes are picked from the table.
- ✓ By using the predefined program it palletize the each component in the palletising box.

Exp No 8 : Machine Tending:



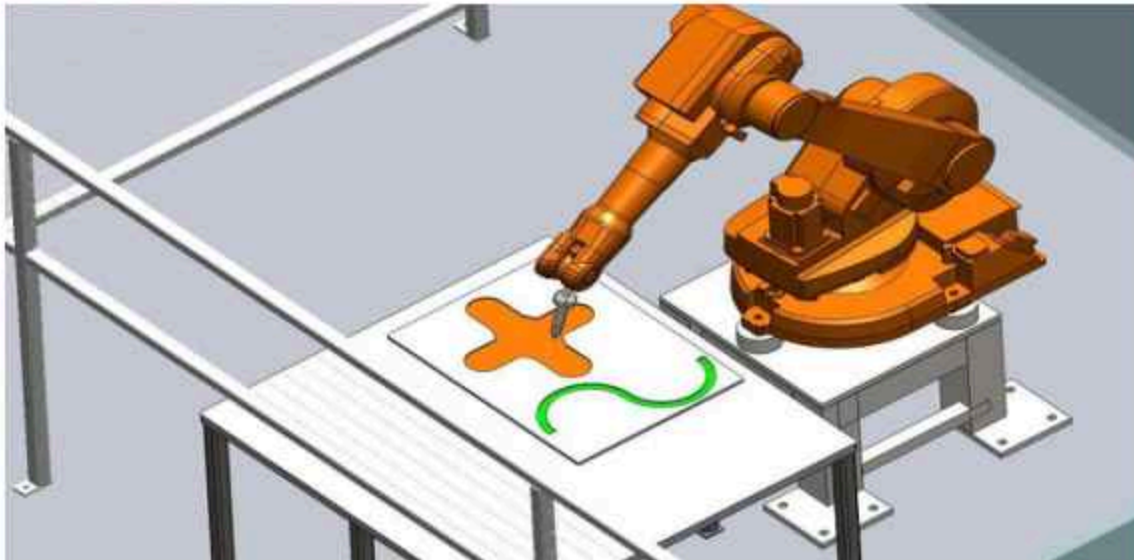
- ✓ The components will be loaded in the pallet placed in the table.
- ✓ The robot will pick the components by using the mechanical gripper and loads it in the pneumatic chuck.
- ✓ The robot waits for some time (assuming as machining time) & then reloads the component.
- ✓ The reloaded component is placed in another pallet in the table.

Exp No 9 :Metal & Non metal components sorting:



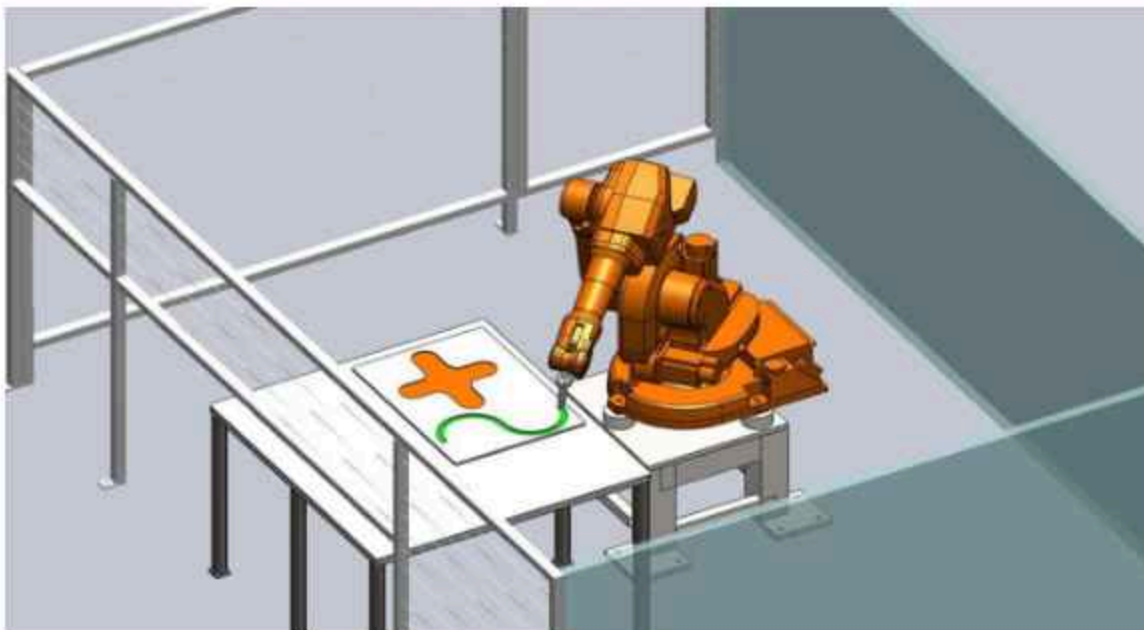
- ✓ Both metallic components & nylon components will be placed in one pallet in the loading table.
- ✓ A mechanical gripper is attached to robotic arm.
- ✓ The robot picks the components from the pallet & moves the component towards the sensor, the sensor detects whether the component is metallic or plastic (nylon).
- ✓ If the component is metallic the robot places those components in one row in other pallet placed in the table.
- ✓ If the component is plastic(nylon) the robot places those components in remaining row of other pallet placed in the table.

Exp No 10 :Simulated Painting:



- ✓ A pre-designed set up will be placed in the table.
- ✓ A dummy painting tool will be attached to the robot arm.
- ✓ As per predefined program the robot is made to paint in the given design.

Exp No 11 :Simulated Deburring:



- ✓ A pre-designed setup will be places in the table.
- ✓ A dummy deburring tool will be attached to robotic arm
- ✓ As per predefined program the robot is made to simulate deburring process in the provided design.

Exp No 12 :Stopping based on sensors :

1. Conveyor start stop based on component sensing under the camera

Exp No 13 :Emergency stop wiring & programming :

1. Emergency button & limit switch links and assigning signals to the robot to stop immediately

Exp No 14 :Explanation & working of robot collision :

1. Robot error message identification & solution.

Exp No 15 :Do's & Dont's :

- ✓ Don't place any load on robot & servo motors
- ✓ Don't unplug the robot cable while the robot is working.
- ✓ Don't interfere during robot in working condition
- ✓ Don't play with robot as it may lead to severe injuries.
- ✓ Proper checking should be mandatory before start of process.
- ✓ While entering in robot cell ensure to hit emergency button & change to manual mode
- ✓ While working with robot ensure the robot is in manual working mode
- ✓ Speed should be about 20% to 30% while working robot in manual mode
- ✓ Place the components in their respective position after completion of experiments.
- ✓ Proper shutdown is mandatory for robot or else it will lead to booting problems.

INDUSTRIAL 6 AXIS Robot (MAKE: MOTOMAN)

Achieve high transfer capability with number-one payload, speed and wrist allowable moment in its class.

- Productivity of customers' equipment can be improved significantly with the robot's high speed (highest speed in 12 kg payload class).
- Acceleration/deceleration control has been improved to achieve maximum reduction of acceleration/deceleration times for all robot postures
- Robot with a 12 kg payload can carry heavy objects and a double gripper can be attached to the arm.

This Robot offers improved acceleration/deceleration control for all robot positions resulting in overall improved productivity. The small footprint, slim arm design allows for minimum installation space and minimizes interference with peripheral devices. Installation is quick and efficient. A single cable is all that is needed to connect the manipulator to the controller, resulting in easy setup and reduced expenses for maintenance and spare parts inventory.

Applications:

Machine Tending, Part Transfer, Assembly

Controller:

YRC1000

Payload:

12 kg

Horz. Reach:

1,440 mm

Vert. Reach:

2,511 mm



INDUSTRIAL 6 AXIS Robot (MAKE: ABB)

The ABB Industrial Robot provides fast and reliable work cycles that boost productivity. Ideal for arc welding application, this robot delivers plenty of performance and a high ROI. The robot features the ability to adjust process speed and position to help achieve optimum manufacturing accuracy with little or no rejects.

The ABB Robot is known for its stiff and robust construction. This translates into low noise levels and long intervals between routine maintenance and long service life. The large working area and long reach are complimented by the compact design and slim wrist. This allows high performance and operation even in difficult and restricted locations.

This Robot will provide a reliable, strong, accurate, and fast arc welding ABB robot. This can use either a IRC5, S4C, or S4Cplus controller.



Robot Information

Robot Specifications		Robot Motion Speed		Robot Motion Range	
Axes:	6	J1	120 °/s (2.09 rad/s)	J1	±170°
Payload:	5.00kg	J2	120 °/s (2.09 rad/s)	J2	±70°
H-Reach:	1444.00mm	J3	120 °/s (2.09 rad/s)	J3	+70° - 65°
Repeatability:	±0.0500mm	J4	280 °/s (4.89 rad/s)	J4	±150°
Robot Mass:	225.00kg	J5	280 °/s (4.89 rad/s)	J5	±115°
Structure:	Articulated	J6	280 °/s (4.89 rad/s)	J6	±300°
Mounting:	Floor				

INDUSTRIAL 6 AXIS Robot (MAKE: FANUC)



Max. load capacity
at wrist: 12 kg



Max. Reach:
1420 mm

Controlled axes	Repeatability (mm)	Mechanical weight (kg)	Motion range (°)						Maximum speed (°/s)						J4 Moment/ Inertia (Nm/kgm ²)	J5 Moment/ Inertia (Nm/kgm ²)	J6 Moment/ Inertia (Nm/kgm ²)
			J1	J2	J3	J4	J5	J6	J1	J2	J3	J4	J5	J6			
6	± 0.08	130	340(360)	250	447	380	380	720	230	225	230	430	430	630	22.0/0.65	22.0/0.65	9.8/0.17



Robot

Robot footprint (mm) 283 x 283

Mounting position Floor •

Mounting position Upside down •

Mounting position Angle •



Controller

30iB

Open air cabinet -

Mate cabinet 0

A-cabinet •

B-cabinet 0

iPendant Touch •

Electrical connections

Voltage 50/60Hz 3phase [V] 380-575

Voltage 50/60Hz 1phase [V] -

Average power consumption [kW] 1

Integrated services

Integrated signals on upper arm In/Out 8/8

Integrated air supply 1

Environment

Acoustic noise level [dB] < 70

Ambient temperature [° C] 0-45

Protection

Body standard/optional IP54 /IP55

Wrist & J3 arm standard/optional IP67



BUDGETARY PROPOSAL

SN	Product Description	QNT	Total Price (INR)
	INDUSTRIAL ROBOT		
1.	Articulated Type Robot (6 Axis) Model: MOTOMAN	1	24,00,000
	ELECTRICAL COMPONENTS:		
1	PLC with Cable	1	
2	Control Panel	1	
3	Gripper(2-Jaw)	1	
4	Vacuum Gripper(Suction Cups)	(1,6 Cups)	
5	Sensor(Proximity &IR)	1	
6	Limit Switch	1	
7	Camera and Cables	1	
8	Other Electrical Accessories	As Required	
			16,00,000
	MECHANICAL COMPONENTS:		
1	Table	2	
2	Bin	2	
3	Pallet	2	
4	Components	1	
5	Conveyor 2m	1	
6	Vision System & Mounting	1	
7	Fencing& Door	1	
8	Pedestal	1	
9	CNC Model Machine with chuck	1	
10	Dummy tool(welding, painting, Deburring)	1	
	TOTAL		40,00,000
	GST@18%		7,20,000
	GRAND TOTAL		47,20,000

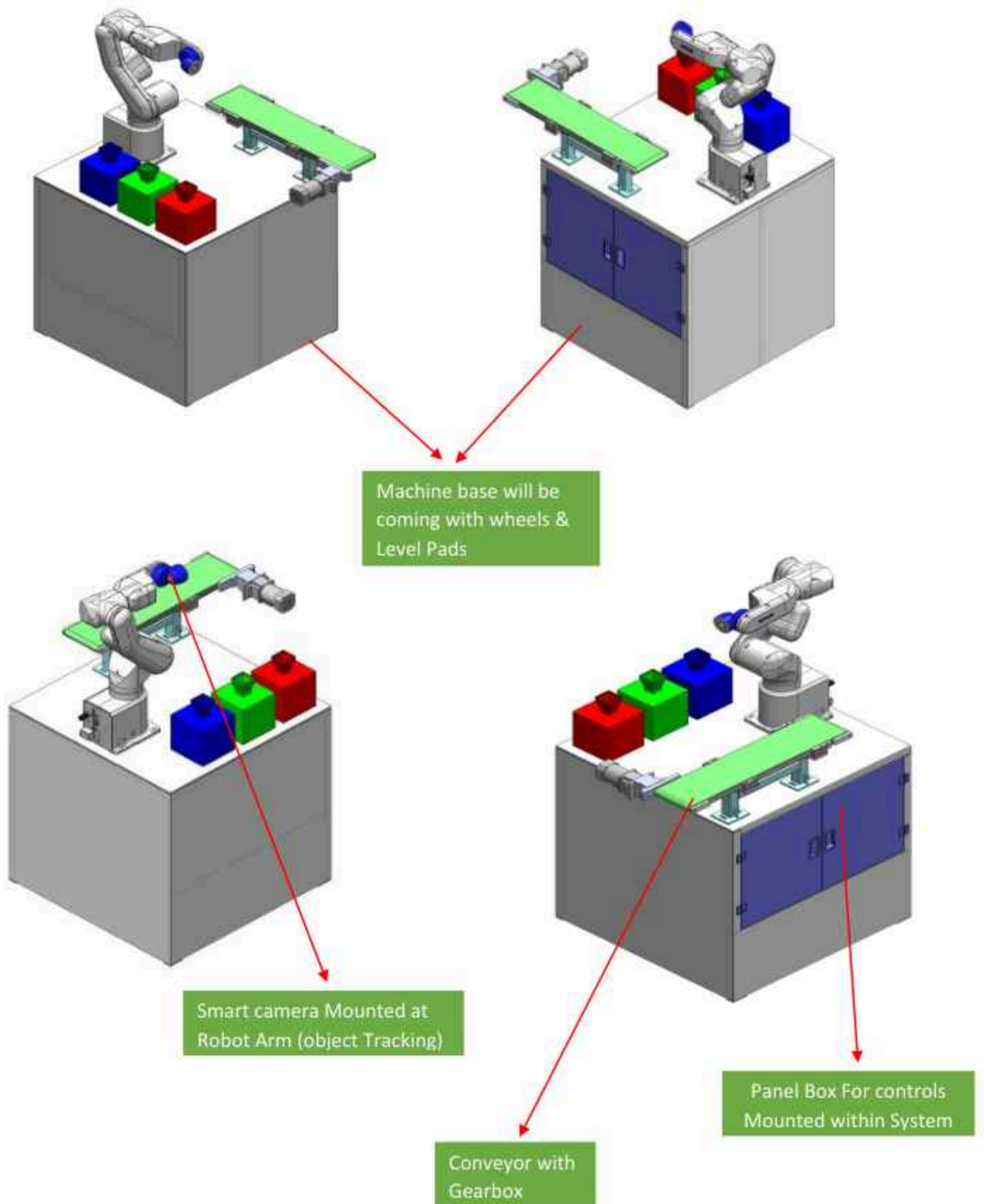
Optional Requirement (You may select Any one of the Robert (MOTOMAN/ABB/FANUC))

SN	Product Description	QNT	Total Price (INR)
	INDUSTRIAL ROBOT		
1	Articulated type Robot (6 Axis) Model: ABB	1	26,00,000
2	Articulated type Robot (6 Axis) Model: FANUC	1	28,50,000

Terms & Condition

- Kindly place your order to:
Emsol Systems,
No: 7, 1st Cross Street (60 Feet Road), B.V.Nagar, Pazhavanthangal
Post, Chennai- 600 114
 - Payment : 90% Advance, 10% After Supply, Installation & Training
 - Tax : GST Extra @ 18 %
 - Delivery : 10 to 12 weeks
 - P& F, Freight : Nil
 - Warranty : 2 years
 - Installation & Training : Free of Cost at your site
 - Validity of Quote : 1 Month
-

System Overview:



SPECIFICATION

<u>Specification Sheet Of Robot</u>	
Arm Length / Reach	From centre axis- 920mm Reach Max
Repeatability:	+/-0.1mm
Payload (Kg):	6kgs
Power:	AC 100 - 240 V single phase Consumption- 1.2kva
I/O's	Standard I/O: In 24/Out 16 & Remote I/O: In 8/Out 8
Axes & Brakes	6Axis has brakes at all axes
Controller	Built-In
Weight	40kgs (max)

Dear Sir,

SUB: Your Requirement of **EPSON Robot** for Robotics Laboratory Setup Application

We are pleased to submit our offer as follow:

QUOTATION OFFER

Robotronix Automation Solution,
"ECM Square" SP- 107, Ambattur
Industrial Estate, Ambattur,
Chennai- 600058



DESCRIPTION	QTY	UNIT PRICE	AMOUNT
Epson VT6L 6-Axis Robot ARM	01	8,60,000/-	8,60,000/-
Vision Sensor (Smart Camera)	01	1,40,000/-	1,40,000/-
Sensors & Switches	-NA-	20,000/-	20,000/-
Conveyor System with Gearbox	01	55,000/-	55,000/-
PLC&HMI	01	45,000/-	45,000/-
Mechanical Systems with Movable & Level Pads	01	65,000/-	65,000/-
Programming & Trainings	-NA-	35000/-	35000/-
Panel Box	01	75,000/-	75,000/-
TOTAL		=	Rs. 12,95,000 /-

GST – Extra @18%

Payment – 30% Advance along with the purchase order and Balance against Proforma Invoice Before Shipment.

Warranty- The above products Shall be warranted against manufacturing Defects for a period of 12 months from the date of Invoice.

Transportation- Charges Will be Seperate

We hope our offer is in line with your requirements and looking forward to the pleasure of receiving your valued purchase order at an early date.

The product will be promised to dispatch within 5-6weeks from the date of payment order received.

Thanking you for the Trust

Yours

Faithfully

For **Robotronix Automation Solution**

Authorised Supporting Companies:-



Sample Submission

Implementation of Robot in an Industry- Micro Assignment

Register Number: 953617114024

Let us assume that you are being the part of Automation division in an industry. Your industry would like to introduce a robot for specific applications like material handling/welding/painting/other processing applications. For this, company give a open bid tender to all suppliers. The quote is attached this assignment (as a separate file). You are the in-charge for procurement of the robot. Compare and answer the following situation and try to justify your selection.

S. No	Criteria	Supplier A	Supplier B	Supplier C
1.	Type of application	Handling Robot	Handling Robot	Handling Robot
2.	Model Name	MOTOMAN (GP12)	ABB	FANUC
3.	No. of axis	6	6	6
4.	Pay load/ Weight carrying capacity	12	5	-
5.	Repeatability	±0.08 mm	±0.05 mm	±0.08 mm
6.	Speed (Angular Velocity)	S-axis 4.53 rad/s, L-axis 4.01 rad/s, U-axis 4.53 rad/s, R-axis 8.20 rad/s, B-axis 8.20 rad/s, T-axis 12.2 rad/s,	J1- 2.09 rad/s, J2- 2.09 rad/s, J3- 2.09 rad/s J4- 4.89 rad/s J5- 4.89 rad/s J6- 4.89 rad/s	J1- 230deg/s, J2- 225deg/s, J3- 230deg/s J4- 430deg/s J5- 430 deg/s J6- 630deg/s
7.	Reach	Horz. Reach: 1,440 mm Vert. Reach: 2,511 mm	Horz. Reach: 1444 mm	From centre axis- 920mm Reach Max
8.	Working Envelope	Spherical	Spherical	Spherical
9.	Configuration type	Vertically Articulated	Articulated	Articulated
10.	Mounting type	Floor-mounted, wall-mounted, Tilt-mounted, ceiling-mounted	Floor-mounted	Floor-mounted
11.	Dimensions	598 (W) x 490 (H) x 427 (D) mm(excluding protrusions)	-	283*283
12.	Weight	130 kg	220 kg	130 kg
13.	Environmental conditions	• Temperature 0° to 45°C • Humidity 20 to 80% RH	-	• Noise level <70 dB • Ambient temperature 0-45

		• Vibration Acceleration Less than 4.9 m/s² (0.5 G) • Free from corrosive gas or liquid, or explosive gas. Free from water, oil, or dust. Free from excessive electrical noise (plasma)		Degree
14.	Electrical requirements	Three-phase 200~240V AC (-15 to +10%) 50/60Hz (±2%)	-	Three phase AC 380 - 575 V 1 kW
15.	Cost	INR 24,00,000	INR 26,00,000	INR 28,50,000
16.	Range of all joints	J1-170°-+170° J2-90°-+155° J3-85°-+150° J4-200°-+200° J5-150°-+150° J6-455°- +455°	J1-±170 J2-±70 J3-+70-65 J4-±150 J5-±115 J6-±300	J1-360 J2-250 J3-447 J4-380 J5-380 J6-720
17.	Others*- T & C (which is also consider for the comparison)	GST 18%	GST 18%	GST 18%
18	Selection	Supplier A robot is selected(MOTOMAN Robot)		

Justification of your selection:

- The MOTOMAN articulated robot has the high payload and high axial movement speed compare to other two.
- The price of this robot is less compare to others.
- Its range of joint movement is also high.
- It have more mounting option and less mass.

S.Ganeshkumar

Signature of the Student